

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomised, Double-blind, Placebo-controlled, Parallel-group Study to Assess the Effect of AZD0780 on Major Adverse Cardiovascular Events in Patients With Established Atherosclerotic Cardiovascular Disease (ASCVD) or at High Risk for a First ASCVD Event [AZURE-Outcomes] **Short Title/ Acronym:** AZURE-Outcomes **Protocol Number:** D7960C00015 **NCT Number:** NCT07000357 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD0780 (Oral PCSK9 inhibitor) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the effect of AZD0780, an oral PCSK9 inhibitor, compared with placebo in reducing the risk of MACE-PLUS in patients with established ASCVD or at high risk for a first ASCVD event. **Secondary Objectives:** To evaluate the time to first event of any component of 3P MACE (cardiovascular death, non-fatal myocardial infarction, non-fatal stroke) and further evaluate safety and tolerability.

2.2 Background & Rationale To address the residual cardiovascular risk in patients with established ASCVD or at high risk for a first ASCVD event despite standard-of-care treatments. Despite available lipid-lowering therapies (like statins), many patients fail to achieve optimal LDL-C levels and remain at high risk for major adverse cardiovascular events (MACE). AZD0780 aims to provide a highly efficacious oral alternative to injectable PCSK9 inhibitors to further lower LDL-C and safely reduce the incidence of cardiovascular events.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Parallel-group (1:1)

3.2 Planned Enrollment & Duration Target **NS:** 15,100 evaluable patients **Number of Sites:** Global multicenter trial **Estimated Study Start:** 04-Jun-2025 **Estimated Primary Completion:** 26-Oct-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Meets one of the following: (a) History of an ASCVD event (ACS 1 to 12 months prior, ischemic stroke 1 to 12 months prior, or revascularization for symptomatic lower limb PAD) OR (b) High risk for a first ASCVD event with LDL-C ≥ 75 mg/dL (≥ 1.9 mmol/L) and at least one additional risk factor (e.g., T2DM, Age ≥ 65 years, non-end stage CKD). 3. Must have achieved a stable background lipid-lowering therapy > 28 days before screening.

4.2 Exclusion Criteria 1. Any underlying known disease, or condition including homozygous familial hypercholesterolemia, or LDL/plasma apheresis within 12 months prior to randomization. 2. Use of PCSK9 inhibitors (evolocumab/alirocumab within 12 weeks, inclisiran within 18 months) or medications like mipomersen, lomitapide, or gemfibrozil. 3. Calculated eGFR < 15 mL/min/1.73 m² at screening, or uncontrolled T2DM defined as HbA1c $\geq 9.5\%$.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: AZD0780 or matched placebo administered once daily. **Route of Administration:** Oral. **Duration of Treatment:** Event-driven study; evaluated from randomization until the primary analysis censoring date (PACD), estimated to take up to approximately 54 months.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background lipid-lowering regimen (e.g., high-intensity statin therapy, or lower-intensity statin combined with an oral agent with proven outcome benefit like ezetimibe and/or bempedoic acid). **Prohibited:** Other PCSK9 inhibitors, mipomersen, lomitapide, and gemfibrozil.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Lipids Labs, evaluate stable SoC therapy
Baseline	Day 0	Randomization, First Dose
Interim	Periodic	Safety Labs, Efficacy Assessment, MACE-PLUS tracking, IP Dispensing
End of Study	After PACD	Study Closure Visit, Final Vital Signs, MACE assessment, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to first occurrence of any component of MACE-PLUS (Major Adverse Cardiovascular Events).

Measurement Method: Clinical event adjudication by an independent Clinical Event Committee (CEC).

7.2 Secondary & Exploratory Endpoints Time to first event of any component of 3P MACE (Cardiovascular death, non-fatal myocardial infarction, non-fatal stroke). Overall safety and tolerability profiles.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via periodic onsite follow-up visits and patient reporting. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. MACE events are captured systematically for primary endpoint assessment. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to review unblinded safety and outcome data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes (MACE-PLUS). Safety Analysis Set for evaluating AEs and SAEs. **Sample Size Justification:** Approximately $N = 15,100$ powered sufficiently for an event-driven trial to accrue the requisite number of MACE-PLUS events. **Handling of Missing Data:** Standard time-to-event methodologies (e.g., Kaplan-Meier estimation, Cox proportional hazards models) with censoring at the date of last contact or the Primary Analysis Censoring Date (PACD).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to assure data integrity and protocol compliance. **Audit Trail:** Data collected in an eCRF system, with query resolution processes and formal adjudication workflows for all suspected cardiovascular events.

11. Study Identifiers & Governance

Primary ID: D7960C00015 **Registry Number:** NCT07000357 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** AZURE-Outcomes **Official Regulatory Title:** A Phase III, Randomised, Double-blind, Placebo-controlled, Parallel-group Study to Assess the Effect of AZD0780 on Major Adverse Cardiovascular Events in Patients With

Established Atherosclerotic Cardiovascular Disease (ASCVD) or at High Risk for a First ASCVD Event

12. Clinical Framework (Cardiovascular Specific)

Primary Condition: Cardiovascular Disease / Atherosclerosis
Diagnosis Threshold: History of ASCVD event or High Risk profile with LDL-C ≥ 75 mg/dL
Medical Methodology: Optimization of standard-of-care background lipid-lowering therapy
Optimization Target: LDL-C reduction and MACE-PLUS risk reduction

13. Study Procedures & Methodology

Management Pathway: Addition of oral PCSK9 inhibitor (AZD0780) to background statin/lipid-lowering therapy
Education Protocol (HCP): Protocol training, MACE event identification, and reporting standards
Education Protocol (Patient): Investigational product compliance and AE/event self-reporting
Quality Audit Mechanism: Independent central adjudication of all suspected cardiovascular events (MACE)

14. Observation & Follow-Up Schedule

Total Duration: Event-driven, up to approximately 54 months
Onsite Visit Cadence: Periodic protocol-defined follow-up visits
Remote Monitoring Frequency: Interim compliance and safety checks
Checklist Review Interval: Continuous tracking through to the Primary Analysis Censoring Date (PACD)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				